

Formal Inverses Are Not Positive-Time Rauzy Dynamics: Exact Cone, Roof, and Trace Obstructions

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Abstract

We test whether identity-holonomy cycles found in a symmetric extension of a Rauzy–AGY symbolic clock can be promoted to genuine periodic dynamics of the positive Rauzy natural extension. The answer is negative for the three source-locked witnesses under consideration. We separate the genuine length action $B^{-\top}$, the contravariant transfer action $B^{\top}$, and a raw covariant homology control. Exact integer Farkas certificates then show that two length-six words and one length-twenty-four relation fail the positive-cone test in every cyclic phase, for both the length and transfer conventions, although their chronological raw products are exactly the identity. A real groupoid one-cocycle is antisymmetric under inversion, so an intrinsic roof cannot be strictly positive on both orientations. Matrix identity is distinguished from the groupoid unit and from vanishing of an arbitrary edge cocycle; a projective normalization cocycle vanishes only conditionally, by telescoping on a domain where every prefix is defined. On the operator side, faithful bounded inverse branches on one infinite-dimensional space obstruct compactness and nuclearity of the non-backtracking block operator. Enlarging the geometric domain instead produces a full neutral fixed family, outside the standard isolated hyperbolic flat-trace atom. The finite group-trace Hashimoto determinant from the preceding calculation nevertheless remains valid as a combinatorial graph object. Thus the result is an obstruction to dynamical promotion, not a retraction of the finite algebraic signal. A strictly positive symmetric edge clock is possible, but it defines a new suspension in which formal inverse labels are symbols rather than inverse arrows of the original dynamical groupoid.

1 Introduction

Transfer-operator constructions often benefit from passing from a one-sided symbolic dynamics to a two-sided or groupoid language. In a hyperbolic system, past and future coordinates can encode stable and unstable directions, and periodic-orbit trace formulae may be organized by primitive cycles and their repetitions. The passage is not formal, however. A symbol denoted e^{-1} can mean at least three different things: the inverse of an arrow in a path groupoid, a branch label in a natural extension, or merely the oppositely oriented edge of a finite graph. Those meanings impose different positivity, time, and trace requirements.

This note examines that distinction in a concrete source-locked experiment. Two length-six cyclically non-backtracking words and a length-twenty-four relation have exact identity holonomy in frozen Rauzy–AGY matrices. In a finite symmetric graph they yield nonzero coefficients in a normalized finite-group trace determinant. That is an interesting algebraic phenomenon: the group trace detects closed words whose chronological cocycle product lies in the kernel. The question is whether the same words can be interpreted as positive-time periodic orbits of the original Rauzy–Veech/AGY dynamics.

The main conclusion is a sharp boundary statement.

For the three certified words, identity of the chronological matrix product is not sufficient for a positive Rauzy orbit, a positive two-sided groupoid roof, or an ordinary same-space

nuclear transfer operator. The finite group-trace determinant survives, but only as the determinant germ of a newly declared symmetric graph system.

There are four logically independent gates behind this conclusion.

- (G1) *Cone gate.* The length vector of a forward Rauzy path evolves by $B^{-\top}$, while a transfer inverse branch is naturally contravariant and uses $B^\top$ in reversed application order. Exact rational feasibility, not a floating-point orbit search, is required for each cyclic phase.
- (G2) *Roof gate.* An additive real cocycle on a groupoid satisfies $\tau(g^{-1}) = -\tau(g)$. Hence a strictly positive assignment to both orientations is not an intrinsic groupoid time.
- (G3) *Operator gate.* If both orientations act as faithful bounded inverses on one infinite-dimensional Banach space, the associated finite non-backtracking block cannot be compact or nuclear. Modern anisotropic and microlocal theories avoid this premise; they do not invalidate the theorem.
- (G4) *Flat-trace gate.* On an enlarged domain, identity projective holonomy gives the identity return map, a neutral fixed family rather than an isolated hyperbolic periodic point.

The cone gate is completely finite and exact. Let chronological products use the convention that later events multiply on the left. The two length-six words are

$$\begin{aligned} C_1 &= (0t, 1b, 0t^{-1}, 0b, 3t, 0b^{-1}), \\ C_2 &= (4t, 6b^{-1}, 6t, 5b, 6t^{-1}, 6b). \end{aligned}$$

The third word, denoted W_{24} , is the frozen length-twenty-four expansion of the earlier A, B, C relation. Exact replay gives

$$B_{C_1} = B_{C_2} = B_{W_{24}} = I_4.$$

Nevertheless, all $6 + 6 + 24$ cyclic phases fail the positive length-cone gate, and all $6 + 6 + 24$ cyclic phases separately fail the transfer gate. The proof uses only integer row certificates: either a required prefix-coordinate row is nonpositive, or positive integer combinations of required rows sum to zero. Two raw covariant homology witnesses remain positive for C_1 and C_2 , which is a useful control that the length and covariant conventions were not silently exchanged.

The conceptual distinction can be summarized as follows.

Object	Meaning of e^{-1}	Consequence for time and trace
Rauzy/AGY positive branch	Exact inverse matrix tested against the source positive cone	May fail to be a legal branch; the AGY return time is positive only on its actual domain.
Path groupoid	Categorical inverse of e	Additive time is antisymmetric; ee^{-1} is a unit of time zero.
Symmetric graph shift	A second oriented edge symbol	A positive symmetric edge length is legal, but defines a new non-backtracking suspension.
Finite group-trace operator	Group-labelled finite edge matrix	Identity holonomy contributes through the group trace without ordinary compactness on $\ell^2(G)$.

This paper is deliberately an obstruction paper. It does not claim that two-sided symbolic dynamics, anisotropic spaces, or group von Neumann traces are generally unsuitable for dynamical zeta functions. Positive-roof suspension theory is classical [10], and anisotropic and microlocal constructions are powerful precisely when genuine hyperbolicity supplies the required contraction and wavefront geometry [3, 5–7]. The local result here says that the certified kernel words cannot be inserted into those theories by identifying formal inverse edges with positive-time AGY inverse dynamics.

The argument is organized as follows. Section 2 fixes chronology and separates the three matrix actions. Section 3 gives the exact all-phase cone certificates. Section 4 proves the groupoid roof obstruction and tracks repetitions. Section 5 gives the same-space nuclearity and flat-trace dichotomy. Section 6 isolates the finite group-trace object that remains valid. Section 7 records the resulting ROUTE–A decision and the nonduplicative next gate. The appendices specify the reproducible certificate protocol and collect the scope boundaries in a single table.

2 Chronology and the three matrix actions

2.1 Source convention

We work with a finite directed Rauzy-labelled graph. An oriented token e has source and target $s(e), t(e)$, an integral invertible matrix $B(e)$, and frozen change-of-frame matrices S_v . A formal inverse token has reversed endpoints and exact matrix

$$B(e^{-1}) = B(e)^{-1}.$$

For a path word $w = e_1 \cdots e_n$, chronological composition is

$$B_w = B(e_n) \cdots B(e_1). \quad (1)$$

Thus later events multiply on the left. The fixed-frame cocycle is

$$g_e = S_{t(e)}^{-1} B(e) S_{s(e)}. \quad (2)$$

This is also the definition for inverse tokens. The following elementary telescoping identity identifies exactly what the finite-group calculation knows.

Proposition 2.1 (Closed-path frame telescoping). *If $w = e_1 \cdots e_n$ is a closed mixed path based at v , then*

$$g_w = g_{e_n} \cdots g_{e_1} = S_v^{-1} B_w S_v.$$

Consequently, for a closed path, $g_w = I$ if and only if $B_w = I$.

Proof. Substitute (2). At every intermediate vertex the adjacent factors cancel because $t(e_k) = s(e_{k+1})$. Closedness identifies the two surviving endpoint frames. Conjugation preserves the identity. $\square$

The closed-path hypothesis is essential. For an open path the endpoint frames differ, and $g_w = I$ need not be equivalent to $B_w = I$.

2.2 Forward lengths

For Rauzy induction, the source convention for the length coordinate is the inverse transpose of the incidence matrix. In the notation of Avila et al. [1, Sections 3.1.3–3.3], a legal positive path has

$$\lambda_k = B(e_k)^{-\top} \lambda_{k-1}. \quad (3)$$

The source positive cone is not optional: a word describes a branch only when one initial vector has every required prefix in that cone. A positive token a uses $B(a)^{-\top}$, while a formal inverse token a^{-1} uses

$$B(a^{-1})^{-\top} = B(a)^{\top}.$$

For a chosen cyclic rotation $e_1 \cdots e_n$, define

$$P_0 = I, \quad P_k = B(e_k)^{-\top} P_{k-1}. \quad (4)$$

The strict length-cone feasibility question is

$$\exists x \in \mathbb{R}_{>0}^4 \quad \text{such that} \quad P_k x \in \mathbb{R}_{>0}^4 \quad (1 \leq k \leq n). \quad (5)$$

For a closed identity word, $P_n = I$; this endpoint fact says nothing about the intermediate inequalities.

2.3 Contravariant transfer branches

A transfer or inverse-branch convention has the opposite application order. For the same raw rotation $e_1 \cdots e_n$, put

$$u_1 \cdots u_n = e_n \cdots e_1,$$

and define

$$Q_0 = I, \quad Q_k = B(u_k)^{\top} Q_{k-1}. \quad (6)$$

The corresponding strict feasibility gate is

$$\exists x \in \mathbb{R}_{>0}^4 \quad \text{such that} \quad Q_k x \in \mathbb{R}_{>0}^4 \quad (1 \leq k \leq n). \quad (7)$$

Equations (4) and (6) must be tested separately. Merely transposing each matrix without reversing chronology tests a third object.

2.4 Raw covariant homology control

The experiment also retains the recurrence

$$H_0 = I, \quad H_k = B(e_k) H_{k-1}. \quad (8)$$

This is a raw covariant action on homology or cocycle coordinates. It is not the AGY length evolution (3) and is not the transfer action (6). A positive vector for (8) is therefore a convention sentinel, not evidence of a Rauzy-length orbit.

2.5 The three frozen witnesses

The source-locked words are

$$C_1 = (0t, 1b, 0t^{-1}, 0b, 3t, 0b^{-1}), \quad (9)$$

$$C_2 = (4t, 6b^{-1}, 6t, 5b, 6t^{-1}, 6b), \quad (10)$$

and a length-twenty-four word W_{24} obtained by chronological expansion of the frozen A, B, C relation. Exact integer multiplication, performed both in the raw arrow alphabet and directly in the A, B, C alphabet where applicable, yields

$$B_{C_1} = B_{C_2} = B_{W_{24}} = I_4. \quad (11)$$

Equation (11) is a source-level statement, not a finite-field accident. It is nevertheless only a holonomy statement. A kernel word of the matrix representation need not be the unit arrow of the free path groupoid, and an identity endpoint does not certify any positive prefix domain.

3 Exact all-phase cone obstruction

The feasibility problems (5) and (7) are homogeneous systems of strict rational inequalities. We avoid numerical optimization entirely. Every failure is recorded by one of two short integer certificates.

Definition 3.1 (Certificate types). Let $\mathcal{R}$ be the finite collection of all coordinate rows of all required prefix matrices in a given phase.

- (i) A *nonpositive-row certificate* is a row $r \in \mathcal{R} \setminus \{0\}$ with $r_j \leq 0$ for every j .
- (ii) A *positive-row-combination certificate* consists of rows $r_1, \dots, r_m \in \mathcal{R}$ and integers $c_1, \dots, c_m > 0$ satisfying

$$\sum_{j=1}^m c_j r_j = 0.$$

Lemma 3.2 (Exact infeasibility). *Either certificate in Definition 3.1 proves that no $x \in \mathbb{R}_{>0}^4$ makes every required row strictly positive.*

Proof. For type (i), $r \cdot x \leq 0$ for every $x > 0$, contradicting the required strict inequality. For type (ii), if each $r_j \cdot x > 0$, then $\sum_j c_j (r_j \cdot x) > 0$, whereas the displayed row identity makes the same quantity zero. $\square$

This is the elementary strict-cone form of a Farkas alternative. Because the matrices, rows, and coefficients are integral, the conclusion has no tolerance or rounding parameter.

3.1 The first length-six word

Theorem 3.3 (All phases of C_1 fail). *For the word C_1 in (9), all six cyclic phases are infeasible for the forward-length action (4), and all six cyclic phases are infeasible for the transfer action (6).*

Proof. In forward-length phase zero, the fourth coordinate of the first prefix and the second coordinate of the fourth prefix have rows

$$r = (0, -1, 0, 1), \quad -r = (0, 1, 0, -1). \tag{12}$$

Both dot products would have to be strictly positive on a feasible x , but their sum is zero. This is a type-(ii) certificate with coefficients 1, 1. For each of the other five rotations, exact chronological replay produces a certificate of one of the two types in Definition 3.1. Reversing each raw rotation and replaying (6) gives six further integer certificates. The full descriptors, including step and coordinate, are generated by the protocol in Appendix A; each can be checked by integer matrix multiplication and a row equality. Hence all twelve strict systems are infeasible. $\square$

The explicit pair (12) is not a proxy for the other phases. Cyclic rotation changes the collection of prefix inequalities, so the all-phase conclusion relies on six separately reconstructed certificates for each action.

3.2 The second length-six word

Theorem 3.4 (All phases of C_2 fail). *For the word C_2 in (10), all six cyclic phases are infeasible for the forward-length action, and all six cyclic phases are infeasible for the transfer action.*

Proof. In forward-length phase zero, the third coordinate row at step three is

$$(-1, 0, 0, 0). \quad (13)$$

It is strictly negative on every $x \in \mathbb{R}_{>0}^4$, so this phase has a type-(i) certificate. The remaining five length rotations and the six contravariant transfer rotations each have an independently reconstructed integer certificate. Applying Lemma 3.2 to the twelve descriptors proves the claim. $\square$

3.3 The length-twenty-four relation

Theorem 3.5 (All phases of W_{24} fail). *Every one of the twenty-four cyclic phases of W_{24} is infeasible for the forward-length action, and every one is infeasible for the transfer action. For each action, the canonical certificate census consists of fifteen nonpositive-row certificates and nine positive-dependence certificates.*

Proof. For the forward-length action, phase zero has at step eight the required second-coordinate row

$$(-11430, -460520, -3353, -456200), \quad (14)$$

which is strictly negative on the positive orthant. Exact phase enumeration then gives fifteen type-(i) and nine type-(ii) certificates.

For the transfer action, phase zero begins, in application order, with A^{-1}, C, B^{-1} . At the third prefix, its second-coordinate row is

$$(-984333, -498163, -999116, -479060). \quad (15)$$

This is again a type-(i) certificate. The other twenty-three phases have exact descriptors reconstructed after reversing their raw path rotations. Their canonical census is likewise fifteen type-(i) and nine type-(ii) certificates. All computations use integer inverses already certified to be unimodular; no floating-point feasibility solver enters. Applying Lemma 3.2 phase by phase proves the result. $\square$

Every phase still has endpoint product I_4 . Thus Theorem 3.5 does not destroy the relation; it shows that no choice of cyclic basepoint turns that relation into a positive-domain orbit under either of the two tested dynamical conventions.

3.4 A control that distinguishes covariant homology

Proposition 3.6 (Raw homology controls). *For the raw covariant recurrence (8), the initial vectors*

$$x_{C_1} = (1, 2, 1, 1)^T, \quad x_{C_2} = (1, 1, 3, 1)^T \quad (16)$$

have strictly positive coordinates at every prefix of C_1 and C_2 , respectively. Shifting the corresponding closed integer trajectories supplies a positive witness in every cyclic phase. The analogous raw covariant feasibility problem for W_{24} is infeasible.

Proof. Multiply the exact chronological matrices successively according to (8) and evaluate them on the two integer vectors in (16). Every displayed coordinate is a positive integer, and the final vector equals the initial one. A cyclic phase begins at the corresponding point of this closed trajectory, proving phasewise positivity. The W_{24} failure has an exact row certificate of the same form as Definition 3.1. $\square$

Remark 3.7 (What the control proves). The first statement of Proposition 3.6 does not rescue either word as an AGY orbit. Its role is adversarial: the same word can pass a covariant positivity calculation while failing both the genuine length and transfer calculations. This detects precisely the transpose/order mistake that would otherwise turn a homology zigzag into a false length orbit.

Corollary 3.8 (Formal-cycle boundary). *The three words are exact identity-holonomy cycles of the finite symmetric path model, but they are not positive-domain periodic orbits for either frozen Rauzy/AGY action tested above. Identity holonomy alone is therefore insufficient to promote their finite trace-log contribution to geometric AGY periodic data.*

4 Groupoid roofs, projective normalization, and repetitions

The cone obstruction already rules out the proposed words on the frozen positive domain. There is a second, more structural obstruction to declaring both orientations to have positive intrinsic time. It follows directly from the cocycle axioms and is independent of the particular matrices.

4.1 Inverse arrows have opposite additive time

Let $\mathcal{G} \rightrightarrows X$ be a groupoid. We write composable arrows as gh , with h applied first. A real one-cocycle is a function $\tau : \mathcal{G} \rightarrow \mathbb{R}$ such that

$$\tau(gh) = \tau(g) + \tau(h) \quad (17)$$

whenever gh is defined.

Theorem 4.1 (Sign obstruction for a two-sided groupoid roof). *Every real additive groupoid one-cocycle satisfies*

$$\tau(1_x) = 0, \quad \tau(g^{-1}) = -\tau(g). \quad (18)$$

Consequently, if e and e^{-1} are categorical inverses, no such cocycle can obey $\tau(e) > 0$ and $\tau(e^{-1}) > 0$.

Proof. From $1_x 1_x = 1_x$ and additivity, $\tau(1_x) = 2\tau(1_x)$, so $\tau(1_x) = 0$. Since $g^{-1}g = 1_{s(g)}$, additivity gives $\tau(g^{-1}) + \tau(g) = 0$. $\square$

Classical suspension-flow zeta functions instead begin with a positive roof over a chosen base transformation or subshift [10, Chapter 6]. Theorem 4.1 does not contradict that construction. It says that if the base alphabet is simultaneously required to be a groupoid with literal cancellation, its additive arrow cocycle cannot be positive on both members of an inverse pair.

4.2 Three meanings of an identity word

An identity matrix product is often enough for a finite group trace, but it is not enough to conclude either path cancellation or vanishing of every time cocycle.

Proposition 4.2 (Identity trichotomy). *Let w be a closed formal word.*

- (i) *If w is the unit arrow of $\mathcal{G}$, then every additive cocycle on $\mathcal{G}$ vanishes on w .*
- (ii) *If only the represented matrix product B_w equals I , an arbitrary antisymmetric edge-constant cocycle need not vanish on w .*
- (iii) *If the cocycle is the logarithmic projective normalizer of the matrix action and all prefix maps are defined, then $B_w = I$ does force the signed sum to telescope to zero.*

Proof. Part (i) is (18). For (ii), a matrix representation may have a nontrivial kernel. An edge-constant antisymmetric cocycle factors through the signed abelianization of the free path groupoid, not necessarily through the matrix quotient. For the present words those abelianizations are

$$[C_1]_{ab} = 1b + 3t, \quad [C_2]_{ab} = 4t + 5b. \quad (19)$$

Choosing a nonzero value on one surviving generator proves the assertion. Part (iii) is established explicitly in Proposition 4.3. $\square$

The length-twenty-four word has zero signed count in each of the frozen A, B, C generators. Therefore every antisymmetric edge-constant cocycle in that three-generator alphabet vanishes on W_{24} . This is a property of that particular signed word; it does not identify the path with the groupoid unit.

4.3 Conditional telescoping of the projective normalizer

Let

$$\ell(x) = x_1 + x_2 + x_3 + x_4.$$

On any cone domain where Mx has positive ℓ -mass, set

$$h_M(x) = \frac{Mx}{\ell(Mx)}, \quad r_M(x) = \log \frac{\ell(Mx)}{\ell(x)}. \quad (20)$$

On the normalized simplex $\ell(x) = 1$, this reduces to $r_M(x) = \log \ell(Mx)$. The domain qualification in (20) is essential: it is precisely what fails for the frozen words on the positive cone.

Proposition 4.3 (Projective normalizer telescoping). *Where the indicated maps are defined,*

$$r_{MN}(x) = r_N(x) + r_M(h_Nx), \quad (21)$$

$$r_{M^{-1}}(h_Mx) = -r_M(x). \quad (22)$$

If $w = e_1 \cdots e_n$, $x_k = h_{B(e_k)}x_{k-1}$, every prefix is defined, and $B_w = I$, then

$$\sum_{k=1}^n r_{B(e_k)}(x_{k-1}) = \log \frac{\ell(B_w x_0)}{\ell(x_0)} = 0. \quad (23)$$

Proof. Homogeneity of ℓ gives

$$\ell(Mh_N(x)) = \frac{\ell(MNx)}{\ell(Nx)}.$$

Taking logarithms and accounting for the initial normalization yields (21). Set $N = M^{-1}$, or evaluate at h_Mx , to obtain (22). Iterating (21) gives the first equality in (23); $B_w = I$ gives the second. $\square$

This proposition is conditional, not a repair of the cone obstruction. For C_1, C_2, W_{24} , Theorems 3.3 to 3.5 say that no positive starting point makes all source prefixes legal in any cyclic phase. If one enlarges the domain to force the algebraic telescoping, one has changed the geometric setting and must separately justify the new transfer theory.

4.4 A positive symmetric clock is a different suspension

Corollary 4.4 (The graph-clock alternative). *An assignment*

$$L(e) = L(e^{-1}) > 0 \quad (24)$$

defines a positive roof over the finite non-backtracking edge shift. It cannot at the same time be an additive cocycle on the path groupoid in which $ee^{-1} = 1$.

Proof. The first assertion is the standard suspension construction over a subshift: sum L along the successive symbols. The second follows because $L(e) + L(e^{-1}) = 2L(e) > 0$, whereas an additive groupoid cocycle must take the unit word ee^{-1} to zero by Theorem 4.1. $\square$

Thus (24) is mathematically legitimate, but its inverse label is an oriented graph symbol rather than positive-time inverse motion in the original AGY groupoid. We refer to the resulting object as the *symmetric Hashimoto suspension*. Calling it a natural extension of the AGY return map would erase the distinction proved above.

4.5 Repetitions do not commute with changing clocks

Periodic-orbit zeta functions organize a primitive orbit together with all of its repetitions. For a positive flow period T_γ , the logarithmic expansion has terms e^{-smT_γ}/m ; see, for example, Dyatlov and Zworski [3, Section 2.2]. The identity words expose a singular boundary.

Proposition 4.5 (Repetition dichotomy). *Suppose a projective matrix-normalizer cocycle is defined along an identity-holonomy word w . Then*

$$T(w^m) = mT(w) = 0 \quad (m \geq 1).$$

Consequently a flow-style Euler factor formally contains $(1 - e^{-s \cdot 0})^{-1}$, and the logarithmic repetition series contains $\sum_{m \geq 1} 1/m$. By contrast, under unit symmetric graph length,

$$T(C_1^m) = T(C_2^m) = 6m, \quad T(W_{24}^m) = 24m,$$

and the corresponding small- u Hashimoto trace-log is regular.

Proof. The zero projective period is (23); additivity gives the same conclusion for powers. Substitution into the primitive flow factor and its logarithm gives the divergence. Graph periods are symbolic word lengths, which multiply under repetition. $\square$

One may use either clock when its hypotheses are satisfied. One may not use zero signed projective time to certify the primitive orbit and positive graph length to regularize its repetitions. That hybrid would not be the trace-log of a single dynamical system.

5 Operator ideals and the flat-trace dichotomy

The failure of a common positive cone does not by itself forbid every transfer operator. Hyperbolic dynamics is routinely represented on anisotropic spaces, and stable and unstable branches can be treated by different polarizations. The correct obstruction is narrower: faithful forward and inverse actions cannot both appear as compact blocks on one infinite-dimensional space.

5.1 A same-space nuclearity obstruction

Let X be an infinite-dimensional Banach space and let $E^\pm$ be a finite oriented edge set with a fixed-point-free involution $e \mapsto e^{-1}$. A transition $e \rightarrow f$ is legal when endpoints match and $f \neq e^{-1}$. Assume each oriented edge has at least one legal predecessor. This holds for the two frozen symmetric graphs.

Choose bounded operators $U_e \in \mathcal{B}(X)$. For nonzero transition weights $c_{f,e}$, define the non-backtracking block operator on

$$X^{E^\pm} = \bigoplus_{e \in E^\pm} X$$

by the blocks

$$(\mathbb{B})_{f,e} = \begin{cases} c_{f,e} U_f, & e \rightarrow f \text{ legal,} \\ 0, & \text{otherwise.} \end{cases} \quad (25)$$

Theorem 5.1 (Faithful same-space inverses obstruct compactness). *Suppose*

$$U_{e^{-1}} U_e = I_X \quad (e \in E^\pm). \quad (26)$$

Then the block operator $\mathbb{B}$ in (25) is neither compact nor nuclear. In a Hilbert realization it is, in particular, not trace class.

Proof. Fix $f \in E^\pm$ and choose a legal predecessor e . Let $J_e : X \rightarrow X^{E^\pm}$ be coordinate injection and $P_f : X^{E^\pm} \rightarrow X$ coordinate projection. Then

$$P_f \mathbb{B} J_e = c_{f,e} U_f. \quad (27)$$

Compact and nuclear operators are two-sided operator ideals. If $\mathbb{B}$ belonged to either ideal, (27) would put every U_f in that ideal. Multiplying by the bounded $U_{f^{-1}}$ and using (26) would put I_X in the same ideal. The identity on an infinite-dimensional Banach space is not compact, while every nuclear operator is compact. This is a contradiction. $\square$

The proof also covers two spaces X_+, X_- if compact arrows in both directions compose to the identity on either infinite-dimensional space. Its hypothesis is not satisfied by a construction in which forward and backward branches act between genuinely different anisotropic completions and are not bounded inverse operators on one common realization. This is how the hyperbolic methods of Gouëzel and Liverani [7], the microlocal approach of Dyatlov and Zworski [3], and stable-unstable pinning constructions such as Baladi et al. [2], Fried [4] escape the elementary ideal argument. Their escape requires hyperbolicity or a cross-map geometry; it is not supplied by matrix identity alone.

5.2 Why cancellation in a sum is not a word theorem

It may be tempting to hope that noncompact blocks cancel in the unmarked sum $\mathbb{B}$. A word-resolved determinant claim needs more.

Corollary 5.2 (Marked coefficient isolation). *Let $\mathbb{B}(\mathbf{z})$ be an edge- or occurrence-marked family that is holomorphic in nuclear norm near $\mathbf{z} = 0$. Every multivariate Cauchy coefficient is nuclear. If a layer marking isolates a relation-word coefficient equal to I_X , or to a boundedly invertible multiplication operator on an infinite-dimensional function space, then such a nuclear-norm holomorphic family cannot exist.*

Proof. The Cauchy coefficient is a Bochner integral of the nuclear-valued map around a small polytorus, hence nuclear. Neither I_X nor a boundedly invertible operator on an infinite-dimensional Banach space is compact. Therefore the isolated coefficient cannot be nuclear. $\square$

The corollary is conditional on a marking that genuinely isolates the ordered word. It does not assert that every unmarked formal sum has such an isolation. Its purpose is to close a specific loophole in claims that already require coefficientwise periodic-word semantics.

5.3 Empty source domain or neutral fixed family

Standard flat traces for hyperbolic maps and flows are built from isolated or cleanly controlled fixed data; the familiar local denominator contains the transverse factor $\det(I - Df)$ [3–5]. For the present identity words there is a sharp alternative.

Theorem 5.3 (Positive-domain/flat-trace dichotomy). *Let $W \in \{C_1, C_2, W_{24}\}$.*

- (i) *On the frozen source positive domain, no cyclic phase of W defines the tested complete length or transfer orbit.*
- (ii) *If the projective domain is enlarged so that every prefix is defined, then $B_W = I$ implies*

$$h_W = \text{Id}, \quad Dh_W = I, \quad \det(I - Dh_W) = 0. \quad (28)$$

The fixed set is the entire enlarged domain, not an isolated hyperbolic periodic point.

The same conclusion holds for every power W^m .

Proof. Part (i) is Theorems 3.3 to 3.5. For (ii), the projectivization of I is the identity map. Its derivative is the identity on every tangent space, so the transverse determinant vanishes. Powers retain both the matrix identity and the identity derivative. $\square$

Thus the standard isolated fixed-point atom

$$\frac{w_W(x_W)}{|\det(I - Dh_W(x_W))|} \quad (29)$$

is unavailable for the very words responsible for the finite group-trace signal. This statement is deliberately not a universal no-go theorem for all distributional traces. A clean-fixed-manifold regularization, a quotient that removes neutral directions, or another renormalized trace may be possible under additional hypotheses. Any such construction would define a new zeta prescription and would require its own invariance, repetition, and analytic-continuation theorem; it cannot be inferred from (29).

6 The finite group-trace determinant that survives

The negative geometric conclusion does not erase the preceding finite calculation. This section states exactly what remains and why it is not in conflict with Theorem 5.1.

6.1 A finite edge matrix over a group algebra

Let G be the discrete group generated by the frozen matrix holonomies and let $\lambda : G \rightarrow \mathcal{B}(\ell^2(G))$ be the left regular representation. The canonical group trace is

$$\tau_G \left(\sum_{g \in G} a_g \lambda(g) \right) = a_e \quad (30)$$

on the group algebra and extends to its group von Neumann algebra. Let $\mathcal{H}$ be the finite non-backtracking matrix whose (f, e) block is $x_f \lambda(g_f)$ when $e \rightarrow f$ is legal and zero otherwise. Chronological composition gives

$$(\mathrm{tr}_{E^\pm} \otimes \tau_G)(\mathcal{H}^n) = \sum_{\substack{e_1 \cdots e_n \text{ closed} \\ \text{nonbacktracking}}} x_{e_1} \cdots x_{e_n} \mathbf{1}_{\{g_{e_n} \cdots g_{e_1} = e\}}. \quad (31)$$

Thus the trace selects based closed non-backtracking words with identity holonomy. With unit marking $x_e = u$, write the number of such based words as N_n . On a sufficiently small disk the trace-log germ is

$$D_\infty(u) = \exp \left[- \sum_{n \geq 1} \frac{N_n u^n}{n} \right]. \quad (32)$$

This is the group-trace analogue of a finite Hashimoto determinant; compare the original edge-zeta framework of Hashimoto [8], Stark and Terras [11] and the von Neumann algebra setting in Lenz et al. [9].

6.2 The frozen normalized finite-Weil limit

The preceding phase of the experiment supplied finite Weil representations ρ_p over odd primes and finite twisted Hashimoto matrices $\mathcal{H}_p$. Its normalized trace input is

$$\frac{1}{p^2} \text{Tr } \rho_p(g) \longrightarrow \mathbf{1}_{\{g=e\}} \quad (33)$$

for every holonomy element occurring in a fixed finite moment. Together with the uniform degree bound $\|\mathcal{H}_p\| \leq M$, this defines, on the common zero-free disk $|u| < 1/M$, the normalized determinant root

$$\Delta_p(u) := \exp[p^{-2} \text{Log}_0 \det(I - u\mathcal{H}_p)] = \exp\left[-\sum_{n \geq 1} \frac{u^n}{np^2} \text{Tr}(\mathcal{H}_p^n)\right] \longrightarrow D_\infty(u). \quad (34)$$

Here Log_0 is the holomorphic logarithm fixed by value zero at $u = 0$. The convergence in (34) is coefficientwise and locally uniform. The finite-head limit follows from (33); the geometric bound

$$|p^{-2} \text{Tr}(\mathcal{H}_p^n)| \leq CM^n$$

gives a prime-independent summable tail on every closed disk $|u| \leq r < 1/M$. It also supplies a common zero-free disk and hence a common logarithm branch through $u = 0$. The frozen degree bounds are $M = 3$ for the C25 oriented graph and $M = 5$ for the C26 rose.

For the C25 graph, exact enumeration gives

$$N_6 = 24, \quad [u^6] \log D_\infty(u) = -\frac{24}{6} = -4. \quad (35)$$

For the C26 graph, the relation and its certified inverse class give

$$N_{24} \geq 48, \quad [u^{24}] \log D_\infty(u) = -\frac{N_{24}}{24} \leq -2. \quad (36)$$

Here N_n counts based closed identity-holonomy words, exactly as in (31); it is not a count of positive AGY periodic orbits. The inequality in (36) is a lower-bound statement for the frozen combinatorial census, not an exact primitive-orbit classification.

6.3 Why this does not contradict the operator obstruction

The trace in (31) is the finite edge-matrix trace tensored with τ_G . It is not the ordinary Hilbert-space trace of $\mathcal{H}$ acting on $\ell^2(G)^{E^\pm}$. In particular, $\lambda(g)$ is unitary and, for infinite G , is not compact; nevertheless its group trace is perfectly defined. Theorem 5.1 rules out an ordinary compact or nuclear same-space realization with faithful inverses. It does not rule out a finite matrix over a finite von Neumann algebra equipped with a different trace.

Likewise, (32) should not be renamed an ordinary Fredholm determinant. No positivity statement identifying it with a Fuglede–Kadison determinant is needed here. Its exact and sufficient status is a small- u group-trace exponential that records identity-holonomy words of the symmetric graph.

Proposition 6.1 (Survival with a strict semantic boundary). *Equations (32)–(36) remain valid after the cone and roof obstructions. What fails is only the identification of their counted words with positive-time periodic orbits of the original AGY natural extension.*

Proof. The finite moments depend on legal graph adjacency, chronological group products, and the group trace. None of those algebraic statements requires a positive Rauzy length vector or a groupoid-positive roof. Conversely, the geometric identification would require the cone and roof gates already shown to fail. $\square$

7 Evaluation and the next nonduplicative gate

The purpose of the ROUTE–A protocol is to prevent a striking finite signal from being promoted before its dynamical and analytic carrier exists. For the specific promotion tested in this paper, the evaluation is

$$(A1_{\text{fail}}, A2_{\text{fail}}, A3_{\text{fail}}, A4_{\text{formal hint}}), \quad \boxed{\text{promotion rejected.}} \quad (37)$$

The four labels have precise meanings.

- A1: fail.** The proposed primitive objects do not exist on the tested positive Rauzy/transfer domains: every cyclic phase of all three source-locked words is infeasible. The unit graph clock is a declaration for a new system, not a recovered AGY return time or an arithmetic primitive law.
- A2: fail.** An ordinary nuclear Hashimoto operator cannot carry faithful same-space inverse branches. Enlarging the domain instead produces a neutral identity fixed family, not the standard isolated flat-trace atom. The C29 finite group trace remains valid as a different trace mechanism.
- A3: fail.** Because the dynamical and operator carriers fail first, questions of global analytic continuation, a functional equation, or a prime-orbit counting law for this promotion are not yet meaningful.
- A4: formal hint.** The exact symplectic kernel relations and normalized finite-Weil traces may be useful as a fibre cocycle over a genuinely hyperbolic base. They do not, by themselves, furnish a self-adjoint Hilbert–Pólya operator or a spectral identification with arithmetic zeros.

No Route–B claim is made. In particular, this work neither fits Riemann-zero data nor infers spectral statistics from the coefficients in (35)–(36).

7.1 What not to rebuild

The associated Hénon program already contains a certified hyperbolic local control, denoted H_6 , with mixed-domain polynomial map

$$H_6(q, p) = (1 - 6q^2 - p, q), \quad (38)$$

rational domain margins, and an exact one-step stable–unstable pinning kernel. Analytic pinning and cross-map methods are the natural setting in which stable and unstable variables are separated, as in the primary construction of Baladi et al. [2]; analytic flow zeta methods likewise have a well-developed hyperbolic foundation [4]. Re-deriving generic local pinning machinery around (38) would therefore duplicate an already closed local gate rather than address the obstruction found here.

7.2 The next large gate

A nonduplicative next theorem should keep the hyperbolic base and move the identity relation into a fibre, where trivial fibre holonomy need not make the base return map the identity. One viable target is the following conditional program.

- (P1) Construct all admissible iterated pinning compositions for the certified H_6 mixed domains and prove a common quantitative contraction bound, rather than only a one-step identity.

- (P2) Place the resulting word operators on compatible polarized spaces and prove a summable nuclear-norm or flat-trace majorant uniform in word length.
- (P3) Add the symplectic/finite-Weil cocycle purely in the fibre. A relation word may have identity fibre holonomy while its base orbit remains an isolated hyperbolic periodic point.
- (P4) Establish a common zero-free determinant disk and a repetition theorem before asking for arithmetic continuation or a Hilbert–Pólya interpretation.

This pivot is intentionally a large gate. It attacks the missing all-word composition and determinant-tail theorem rather than extending the failed formal-inverse scan to longer words or more primes. Success would evade both parts of Theorem 5.3: the base word would have a genuine positive period and nonneutral derivative, while identity would occur only in a finite or group-traced fibre.

7.3 Scope of the novelty claim

The mathematically new local content claimed here is limited to the source-locked conjunction of: (i) exact all-phase cone certificates for the three frozen words under both correct action conventions; (ii) their raw covariant controls; and (iii) the resulting roof, repetition, and trace semantic boundary. The general cocycle sign identity, suspension formalism, operator-ideal fact, and hyperbolic trace machinery are standard. We make no claim that every possible natural extension or every regularized groupoid zeta is obstructed.

8 Conclusion

The identity-holonomy signal survives exact scrutiny, but its correct home is narrower than initially hoped. Chronological replay proves that the two length-six cycles and the length-twenty-four relation have raw product I_4 . Exact integer cone certificates simultaneously prove that none of their cyclic phases lies in the tested positive Rauzy length or transfer domains. Positive raw covariant witnesses for the two shorter words show why the separation of $B^{-\top}$, reversed $B^\top$, and B is essential.

At the conceptual level, categorical inversion forces an additive roof to change sign. Matrix identity does not turn a kernel word into a unit arrow, and it kills the projective normalizer only on a domain where the complete chronological path is defined. If such a domain is artificially enlarged, the return map is the identity and its fixed set is fully neutral. If instead one assigns positive symmetric edge length, one obtains a valid and useful non-backtracking graph suspension—but a different dynamical system.

The operator analysis gives the same boundary in another language. Bounded faithful inverse branches on one infinite-dimensional space cannot be compact or nuclear blocks. Anisotropic, microlocal, and pinning theories can escape that theorem only by supplying genuine hyperbolicity and distinct polarizations. The finite group-von-Neumann trace does not need ordinary compactness and therefore retains the exact C29 determinant coefficients.

Accordingly, the tested AGY promotion is rejected at the first two ROUTE–A gates. This is a useful negative result: it closes an entire formal-inverse route without a longer word or prime scan, while preserving the finite algebraic object. The next credible construction should place the identity-holonomy cocycle in a fibre over an independently certified hyperbolic base and prove an all-word determinant-tail theorem. That move changes the dynamical architecture rather than relabelling a failed positive cone.

A Exact certificate and replay protocol

This appendix specifies a finite reconstruction of every computational claim used in Section 3. It makes the result reproducible without treating the producer’s verdict as a proof oracle.

A.1 Frozen inputs and semantic firewall

Six upstream artifacts are locked by full SHA–256 before any matrix is read: the C25, C26, and C29 certificate/theorem pairs. The frozen digests are listed below.

Artifact	SHA–256
C25 certificate	a35cee22714abbb9dc9aadcc165720d1ff77aff3b7f29071f53a1b451760bd12
C25 theorem	e1835d63bef914b355ceb4f64acc9043d11a842e9f4e59c7573c63ff66d03702
C26 certificate	1c0289b9b47e65e0603ea001be7cce263aea13d58c66e4609eac88edf8f7ce4a
C26 theorem	4e882cbc332711b4cd2f98e9530f89268c8fcf1712eb150aacfee968dcf50495
C29 certificate	412840c37d2e474462b39ce7072614323023ac8e3f968bc16a9219cc3a0c0cca
C29 theorem	9b53a2d2971f9b3ac7e43193860345d83bfd2343cbcf4fe94acdd47aed3824cc

The experiment reads only exact integral matrices, path endpoints, formal inverse labels, and the three frozen relation words. It does not read a table of primes, zeta zeros, fitted weights, averaged transition matrices, or words outside the declared witnesses. Chronology is preserved at every step; products are never replaced by an average transition matrix.

A.2 Independent raw-product replay

For each raw path rotation $e_1 \cdots e_n$, reconstruct the symbol matrix by

$$B(e^\varepsilon) = \begin{cases} B(e), & \varepsilon = 1, \\ B(e)^{-1}, & \varepsilon = -1, \end{cases}$$

where the inverse is computed by exact unimodular elimination. Form the later-on-left product

$$B(e_n) \cdots B(e_1).$$

It must equal I_4 for every cyclic rotation. The W_{24} product is also reconstructed from the frozen A, B, C matrices before its arrow expansion is used. This double replay distinguishes a source identity from a finite-field or change-of-frame artifact.

A.3 Phase enumeration and action order

For $w = e_1 \cdots e_n$, define the left rotations

$$R_j(w) = e_{j+1} \cdots e_n e_1 \cdots e_j, \quad 0 \leq j < n.$$

For each $R_j(w) = a_1 \cdots a_n$, perform the following two independent replays.

Length replay.

Use application sequence $a_1, \dots, a_n$ and set

$$P_k = B(a_k)^{-\top} P_{k-1}, \quad P_0 = I.$$

Transfer replay.

Use application sequence $a_n, \dots, a_1$ and set

$$Q_k = B(a_{n+1-k})^\top Q_{k-1}, \quad Q_0 = I.$$

The terminal prefix must be I_4 in both replays. For feasibility, collect the four rows of I_4 at step zero and the four rows of every subsequent prefix, ordered lexicographically by (step, coordinate).

A.4 Canonical Farkas descriptor

Given the ordered row list, apply this deterministic selection rule.

1. Choose the first nonzero row r with $r_i \leq 0$ for all i , if one exists. The machine-readable certificate expands it as

$$r + \sum_{i:r_i < 0} (-r_i) e_i = 0, \quad (39)$$

where the e_i are the step-zero rows. Every coefficient displayed in (39) is a positive integer.

2. Otherwise scan supports of size 2, 3, 4, 5, in lexicographic row order. For each support, solve

$$\sum_j c_j r_j = 0 \quad (40)$$

by exact rational row reduction. Accept the first one-dimensional nullspace whose generator has all coordinates of one sign; orient it positively, clear denominators, and divide by the coefficient gcd.

The support bound five is complete for a minimal positive dependence in $\mathbb{R}^4$. Every accepted descriptor is checked by recomputing the weighted row sum and applying Lemma 3.2. Producer and checker independently reconstruct the matrices, rotations, prefix rows, and primitive coefficients; the checker does not import producer functions.

A.5 Headline rows and census

The short human-checkable phase-zero witnesses are collected in Table 2.

Word/action	Step, coordinate	Row	Certificate
C_1 , length	1, 4 and 4, 2	$(0, -1, 0, 1), (0, 1, 0, -1)$	Opposite pair
C_2 , length	3, 3	$(-1, 0, 0, 0)$	Nonpositive row
W_{24} , length	8, 2	$(-11430, -460520, -3353, -456200)$	Nonpositive row
W_{24} , transfer	3, 2	$(-984333, -498163, -999116, -479060)$	Nonpositive row

Table 2: Selected exact phase-zero rows. Step and coordinate use one-based human indexing; the JSON certificate records the coordinate in its native zero-based form.

The final census is

Word	Number of phases	Length failures	Transfer failures
C_1	6	6	6
C_2	6	6	6
W_{24}	24	24	24

For C_1 and C_2 , the covariant recurrence is replayed separately. The phase-zero trajectories from (16) are positive and close exactly. Every cyclic phase is initialized by the corresponding point on that positive integer trajectory, which proves all cyclic raw-covariant controls without changing the action convention. No such positive raw trajectory exists for W_{24} .

B Claim and scope ledger

Statement	Status	Exact scope
Raw products of C_1, C_2, W_{24} are I_4	Exact theorem	Closed frozen words with later-on-left chronology; not an assertion about arbitrary words.
All cyclic positive-cone gates fail	Exact theorem	The three words, under the frozen forward-length and contravariant transfer actions only.
Raw homology controls remain positive	Exact control	All phases of C_1 and C_2 under B , not under $B^{-\top}$ or reversed $B^{\top}$.
Inverse groupoid time changes sign	General theorem	Every real additive one-cocycle on a genuine groupoid.
Identity holonomy implies arbitrary roof zero	False in general	True for a unit arrow, or conditionally for a cocycle that factors through the matrix action on a common domain.
Symmetric positive edge clock exists	Valid construction	A new finite non-backtracking graph suspension; inverse labels are graph symbols.
Same-space nuclear Hashimoto operator exists	Obstructed	When every edge operator is bounded on one infinite-dimensional space and the opposite edge is its faithful bounded inverse.
All anisotropic transfer theories are obstructed	Not claimed	Different polarizations and non-inverse cross maps fall outside Theorem 5.1.
Standard isolated fixed-point atom exists	Obstructed	For the three identity words on any enlarged domain where the full projective return is the identity.
Every clean fixed-set regularization fails	Open	A new distributional or quotient trace would require separate hypotheses and proofs.
C29 finite group-trace determinant germ survives	Exact imported control	Small- u graph determinant with based identity-holonomy counts and frozen norm disks.
C29 determinant is an ordinary Fredholm determinant	Not claimed	The group trace is not the ordinary trace on $\ell^2(G)$.

Statement	Status	Exact scope
Arithmetic/Hilbert–Pólya realization follows	Rejected	No self-adjoint operator, continuation, functional equation, or zero correspondence is constructed.
Hénon H_6 local pinning must be rebuilt	Duplicate route	Existing mixed-domain and one-step control should instead feed an all-word composition/tail theorem.

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